LSV Analysis: iR Compensation, Tafel Slope

■ **Live app:** https://lsv-analysis-ir-compensation-tafel-slope.streamlit.app/

A Streamlit app for two independent electrochemistry workflows, run from uploaded Excel/CSV files — no coding required:

1. **iR compensation** — correct linear-sweep voltammetry (LSV) data for ohmic drop, using the uncompensated resistance **Ru** fitted from an EIS Nyquist arc.
2. **Tafel slope analysis** — fit the linear (activation-controlled) region of a polarization curve to extract the Tafel slope, with RHE reference conversion and mechanistic interpretation (HER, HOR, OER, ORR, CO■RR, N■RR, NO■RR, and more).

Quick start

```
pip install -r requirements.txt
streamlit run app.py
```

Open the URL shown in the terminal (default `http://localhost:8501`). In the sidebar, pick **Use bundled sample** to try the EIS/LSV tabs immediately with the example workbook in `sample-data/`.

PNG downloads need a headless Chrome (via `kaleido`); if it's missing, run `plotly_get_chrome`, or use the HTML/CSV download options instead.

How to use it

1 · EIS / Ru Analysis and 2 · LSV iR Correction

1. **Upload** an Excel workbook (Sheet 1 = EIS: `Z'`, `Z''`; Sheet 2 = LSV: `Potential`, `Current`) or two CSV files. Several samples can sit side-by-side in one sheet as repeated column pairs; pick the active one from the sidebar.
2. **Confirm units.** Current unit and Ru unit are auto-detected from the column headers. If one is per-area (e.g. mA/cm²) and the other isn't, enter the **electrode area (cm²)** to reconcile them.
3. **EIS / Ru Analysis tab** — Ru is read off the high-frequency intercept of a circle fit to the Nyquist semicircle (adjust the fit range, or enter Ru manually).
4. **LSV iR Correction tab** — pick one or more compensation factors (5–100 %; **85 % recommended**):

`text E_corrected = E_measured − (factor / 100) · I · Ru`

View the corrected curve against the raw one, then download the result as CSV. An over-compensation ("fold-back") guard flags any factor that's unsafe and suggests the highest safe one.

3 · Tafel Slope Analysis

Independent of the tabs above — its own uploader, units, and reference electrode.

1. **Upload** one or more Potential/Current files (Excel or CSV), one per sample; combine them into a single overlay plot.
2. **Convert to RHE** if your data isn't already on that scale: pick the reference electrode used (SCE, Ag/AgCl variants, Hg/HgO, …) and the electrolyte pH.
3. The **linear Tafel region is auto-detected** from the reaction onset; fine-tune it per sample by dragging the slider or box-selecting the region directly on the plot.
4. **Tag each sample's reaction** (HER, HOR, OER, ORR, CO■RR, N■RR, NO■RR, and more) — samples of different reactions can share one overlay, split into per-reaction legend groups.
5. Read off the **Tafel slope, R^2, and nearest mechanistic benchmark**, then download the plots (PNG/HTML) and plotted data (CSV) — publication-styled with Arial fonts and a selectable font size.

Good to know

- **LSV data.** Linear-sweep voltammetry records current as the electrode potential is swept linearly — every tab starts from a `Potential, Current` (or current-density) pair read from your file.
- **Potential → RHE conversion** (Tafel tab). If your data is referenced to another electrode (SCE, Ag/AgCl, …), it's converted with `E(RHE) = E(measured) + E°(ref vs NHE) + 0.0592 V·pH■¹ × pH`. Skip this if your data is already vs RHE.
- **Current & current-density conversion.** Switching the current-unit dropdown (A ↔ mA ↔ µA ↔ nA, or their `/cm²` counterparts) instantly rescales the values within that family. Converting *between* an absolute current and a density needs the **electrode area**: `current density [A/cm²] = current [A] / area [cm²]`. The same area also reconciles `Ru` between Ω and Ω·cm² so `I·Ru` always resolves to volts.
- **Privacy.** Uploaded files are processed in memory only — nothing is written to disk — and a **Clear / reset** button wipes the session. An optional password gate can be set via a `app_password` secret.
- Full technical notes (data format, security hardening, using the Python API without the GUI) are in the module docstrings under `scripts/modules/`.

Citation

If you use this app (including the public Streamlit deployment) in your work, please cite it. Machine-readable metadata is in `CITATION.cff`; a plain-text form:

> Kumar, R. (2026). *LSV Analysis: iR Compensation and Tafel Slope* (v1.1.0) [Computer software]. North Carolina Central University. https://github.com/rajeev4187/LSV-Analysis-iR-compensation-Tafel-slope

```
@software{kumar_lsv_analysis_2026,
 author  = {Kumar, Rajeev},
 title   = {LSV Analysis: iR Compensation and Tafel Slope},
 version = {1.1.0},
 year    = {2026},
 url      = {https://github.com/rajeev4187/LSV-Analysis-iR-compensation-Tafel-slope}
}
```

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